Technical University of Denmark

Bachelor Contracrt

Software for Orientation Analysis Using Structure Tensors in 2D and 3D

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1 Objective

This bachelor project aims to investigate and implement computational methods for analyzing two-dimensional (2D) and potentially three-dimensional (3D) fiber image data using structure tensors. Structure tensors are mathematical tools for extracting local orientation, edge information, and features in images. The project combines theoretical understanding with practical software development, focusing on implementing reliable gradient and tensor computations, eigenvalue analysis, and visualization tools. The primary objective is to create a coherent, well-documented, and extensible software solution that makes working with structure tensors in both 2D and 3D clear, accessible, and technically robust.

2 Timeline

February 04 - February 28

- Investigate and research the use of the Gaussian Kernel to smooth discrete signals with noise in 1D.
- Calculate and test of smoothing with gaussian and finding the gradient using the central difference method
- Smooth the signal and find the gradient simultaneously by using the gaussian derivative.
$$\nabla(G_\sigma * I) = (\nabla G_\sigma) * I$$
- Extend our use case to 2D grayscale images.
- Use the gaussian derivative to smooth 2D images with noise and find the gradients, to be able to detect edges in the image.

March 01 - March 31

- Research Structure Tensors and Eigen decomposition.
- Calculate and test of structure tensors for individual pixels in 2D grayscale pictures.
- Calculate the dominant orientation for individual pixels in 2D grayscale pictures.
- visualize the dominant orientation for individual pixels in a picture as arrows or with colors on top of a picture.
- Optimization of Structure Tensors

April 1 - April 30

- Composition of Software for orientation analysis

- Research existing methods and determine a new possible use.

May 1 - May 31

- Report writing
- write Source, references, and appendix
- presentation preparation
- create slideshow

3 Meetings

For the duration of the project, we plan on having a weekly meeting with our supervisor Vedrana to track progress and get feedback throughout. We try to coordinate for scheduling of meetings to compensate for busy schedules, and cancellations due to absents. We communicate within working hours by email for further questions and follow ups.

4 Risk Considerations

- **Complexity of 3D implementation:** Extending the methods from 2D to 3D increases mathematical and programming complexity.
- **Computational cost:** 2D and 3D tensor computations may require significant processing time and memory.
- **Visualization challenges:** Presenting 3D orientation and structural information in a clear way may be technically demanding.
- **Debugging and numerical stability:** Errors in gradient, tensor, or eigenvalue computations may require additional time to resolve.